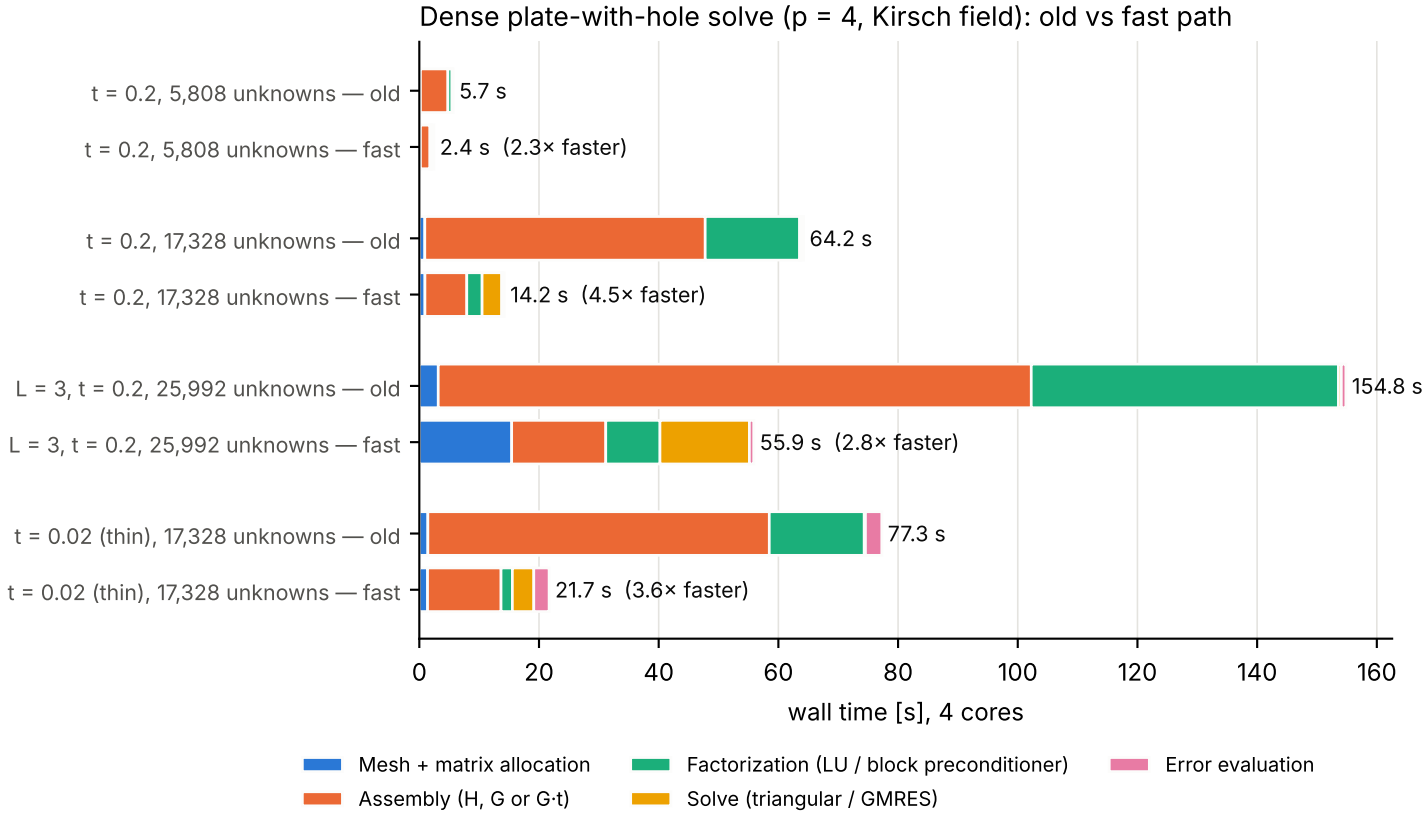
Dense plate-with-hole solve (p = 4, Kirsch field): old vs fast path
t = 0.2, 5,808 unknowns — old
5.7 s
t = 0.2, 5,808 unknowns — fast
2.4 s (2.3× faster)
t = 0.2, 17,328 unknowns — old
64.2 s
t = 0.2, 17,328 unknowns — fast
14.2 s (4.5× faster)
L = 3, t = 0.2, 25,992 unknowns — old
154.8 s
L = 3, t = 0.2, 25,992 unknowns — fast
55.9 s (2.8× faster)
t = 0.02 (thin), 17,328 unknowns — old
77.3 s
t = 0.02 (thin), 17,328 unknowns — fast
21.7 s (3.6× faster)
0
20
40
60
80
100
120
140
160
wall time [s], 4 cores
Mesh + matrix allocation
Assembly (H, G or G·t)
Factorization (LU / block preconditioner)
Solve (triangular / GMRES)
Error evaluation